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The Test of Phased Array Antenna for Marine Seismic Data Transmission

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Abstract

This paper mainly analyzes the test results of a phased array antenna for real-time sea-area seismic datatransmission. The real-time data backhaul effects to the onshore data processing center through the phased array antenna in the sea-area seismic detection process was tested in the specified sea area. The test results show that the phased array antenna can transmit stably the detection data across the sea for 105km. The transmission delay is within 2 to 3 seconds, and the transmission rate reaches 40Kbps while the transmission loss rate is only about 1.5%. 99.5% of the detection data is well received and the power consumption of the whole system is 30W. It can be seen from the test results that the phased array antenna system can satisfactorily meet the real-time seismic databackhaul requirements in the transoceanic seas, and thus can provide technical support for the development of seismic detection

Keywords: Phased array antenna, sea-area seismic detection, real-time data backhaul

1. Introduction

Most of the earthquakes in the world take place on the seabed. The internal structure model of the Earth obtained through the continental seismic observation system can no longer express the details of the mantle in the ocean and the seismic mechanism of the submarine earthquake(1). Therefore, high-quality long-term seismic observations in marine areas turn out to be an important part of the global seismic observation network(2). In the

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past 50 years, European countries, the United States and Japan have continued to promote the development of marine seismic observations.

At present, global oceanic seismic observation has entered the long-term fixed-point real-time observation stage of the deep seabed. Several Pacific Rim countries such as the United States, Canada and Japan have already established large-scale submarine seismic observation networks on the seabed(3). Due to historical reasons, the development of marine science and technology in China lags behind that of West developed countries. However, this emerging technology of the submarine seismic observation network provides a valuable opportunity for developing China's marine science and technology. China's submarine seismic observation research dates back to the "863 Plan". However, the existing submarine seismic observation network generally uses the submarine optical cable to connect the connection box for real-time submarine seismic data transmission. Such kind of transmission method costs highly, and is difficult to construct.

This paper mainly introduces another communication mode that is different from the one using the submarine cable connection, and this mode adopts a phased array antenna system for wireless transmission. Compared with the traditional mechanical scanning antenna, the phased array antenna has the advantages of more flexible beam scanning, a higher beam switching speed and wider beam scanning space, so it can better meet the requirements of modern wireless communication, and its application prospect is very broad. Theoretical analysis and experimental comparison of the transmission system have been conducted. Through the field test, the feasibility of the phased array antenna wireless data transmission system for submarine seismic observation data backhaul is preliminarily proved, which guarantees the communication for the marine seismic network.

2. Phased array antenna wireless data transmission system

The phased array antenna wireless data transmission system contains a maritime wireless transmitting terminal and an onshore wireless receiving terminal. The maritime transmitting terminal directly connects the seismic sensor through the transmission line, and once the sensor collects the data, it will transmit the data back to the onshore receiving terminal through the phased array antenna. As is shown in Fig 1. The phased array antenna system acquires the real-time communication information between the sender and the receiver through theintelligent solution algorithm, and establishes intelligentlythe optimal communication channel, which improves the radio spectrumutilization rate and signal transmission efficiency.

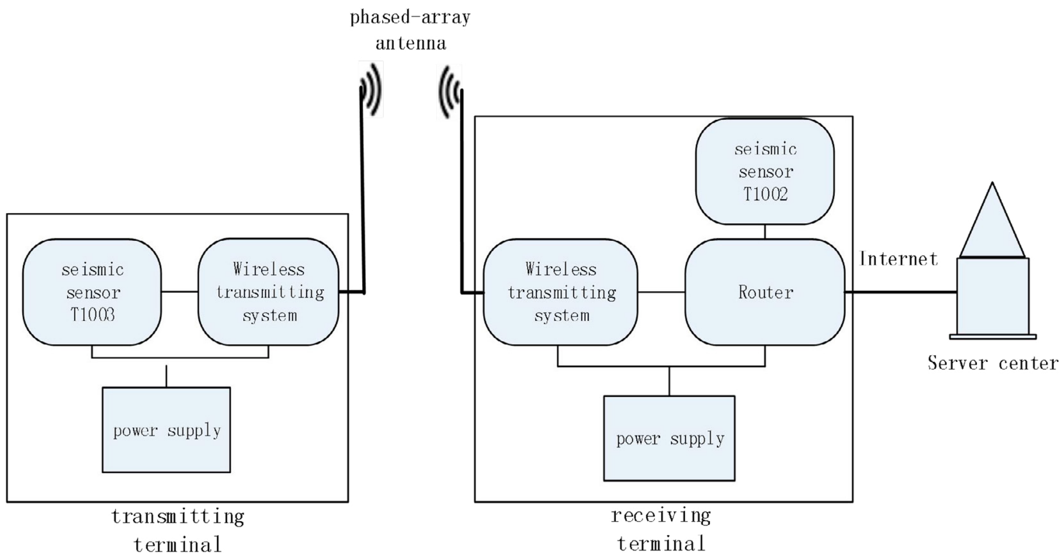

Fig 1. The schematic diagram of the phased array antenna wireless data transmission system

As is shown in Fig 2, Assuming that the antenna heights are h_1 and h_2 respectively, Due to the Earth's surface curvatureand, then the geometric visual distance D is[4]:

$$D = \sqrt{2}R(\sqrt{h_1} + \sqrt{h_2}) \tag{1}$$

Since the earth radius R is about 6370km, the equation above can be simplified to be:

$$D = 4.12(\sqrt{h_1} + \sqrt{h_2}) \tag{2}$$

Assuming that the antenna height at the maritime transmitting terminal is about 15m and that the altitude of the onshore receiving base station is 420m, the transmission distance of the phased array antennawireless data transmission system can be calculated to be about 100 km.

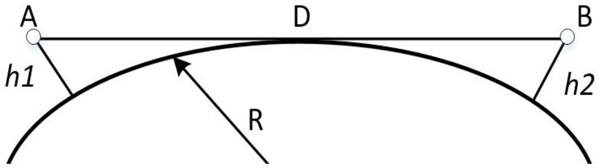

Fig 2. The schematic diagram of transmission distance

Reference the Egli mode, the maritime wave transmission loss can be roughly calculated by[5]:

$$L_g = 88 + 20\lg f + 40\lg d - 20\lg h_1 h_2 - K_s \tag{3}$$

in which L_g is the transmission loss, its unit is dB. f signifies the communication frequency. d is the communication distanc., h_1, h_2 are respectively altitudes of the transmitting and receiving antenna, and K_s is the terrain correction factor. Since the sea level is close to a regular plane, the value of K_s is taken zero, and then the formula above can be reduced to be:

$$L_g = 88 + 20\lg f + 40\lg d - 20\lg h_1 h_2 \tag{4}$$

the transmission loss is related to many factors such as frequency, the distance, and heights of the transmitting and receiving antennas.

3. Test result analysis

According to the analysisabove, a sea area is selected to conduct the real-time transmission test of sea-crossing long-distance submarine seismic data. The specific test results are presented as follows:

3.1. Real-time waveform

It can be seen from the Fig 3 that after the sea-crossing transmission of the wireless data transmission system, the seismic sensor T1003 at the maritime transmitting terminal and the seismic sensor T1002 at the onshore receiving base stationcan both return the detected data to the shore-based data receiving center in real-time.

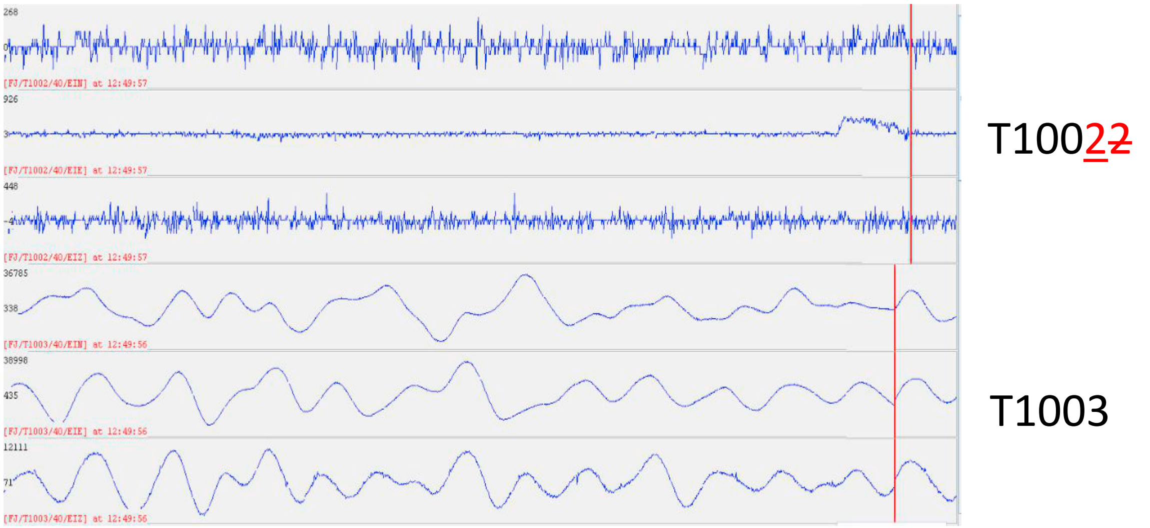

Fig. 3. The real-time waveform

3.2. Transmission delay

The Fig 4signifies that after sea-crossing transmission, the data transmitted by the maritime transmitting terminal has an average transmission delay between 3s and 5s which is 1s to 2s more than that of the data transmitted by the shore-based seismic sensor T1002.

	avg/ms		max/ms	min/ms
FJ/T1002	2135	631.44	3197	1117
FJ/T1002	2131	637.65	3194	1115
FJ/T1002	2154	634.39	3196	1121
FJ/T1002	2238	652.29	3512	1177
FJ/T1002	2314	700.72	3684	1127
FJ/T1002	2370	713.94	3683	1120
FJ/T1002	2450	702.94	3696	1178
FJ/T1002	2407	714.14	3690	1127
FJ/T1002	2433	713.24	3696	1123
FJ/T1002	2448	710.49	3693	1175
FJ/T1002	2403	754.03	4187	1137
FJ/T1002	2358	703.32	3683	1119
FJ/T1002	2377	715.14	3687	1163
FJ/T1003	3496	931.72	8075	1862
FJ/T1003	3691	1153.57	8502	1358
FJ/T1003	3740	1145.82	7569	1560
FJ/T1003	3618	939.91	7070	1830
FJ/T1003	3892	1114.94	8565	2058
FJ/T1003	3602	1042.14	8323	1898
FJ/T1003	3487	946.44	7215	1822
FJ/T1003	3827	1229.38	9862	2251
FJ/T1003	3873	1265.55	9867	1597
FJ/T1003	3850	822.28	6214	2249
FJ/T1003	3514	702.48	5467	2059
FJ/T1003	3729	761.18	5364	1678

Fig. 4. The transmission delay

3.3. Transmission rate

According to the background control system's monitoring, when the transmission baud rate is 19200 Kbps, the transmission rate of the seismic sensor T1003 at the maritime transmitting terminal is 5-15 Kbps and it can reach 30-40 Kbps when the data is replenished. As is shown inFig 5.

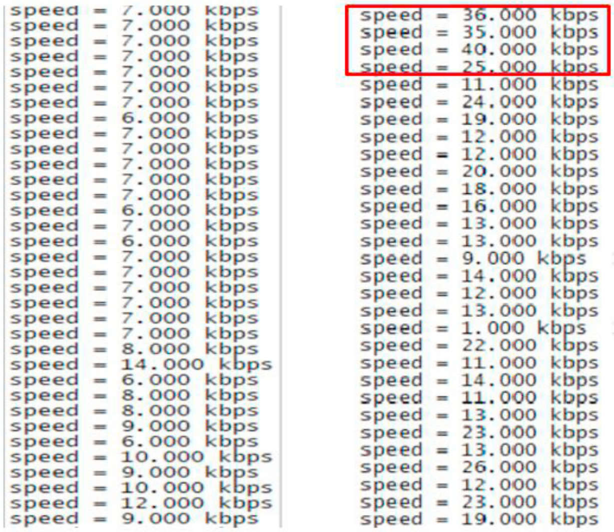

Fig. 5. The transmission rate

3.4. Transmission loss rate

When the transmission baud rate is 19200 Kbps, the transmission loss rate of the seismic sensor T1003 at the maritime transmitting terminal is about 1.5%.As is shown inFig 6.

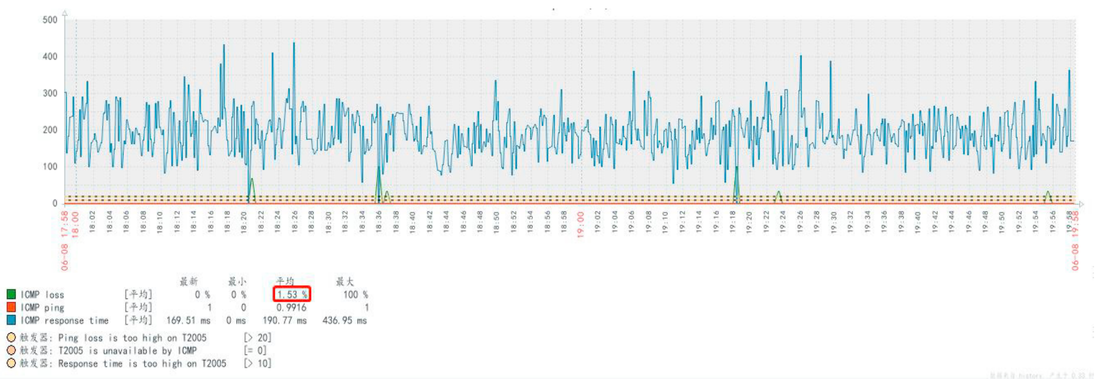

Fig. 6. The transmission loss rate

3.5. Data receiving rate

Since the data receiving system is provided with the replenishment function when packet loss occurs, the final datareceived by the whole system can reach more than 99.7%, which meets therequirement of data receiving validity. As is shown in Fig 7.

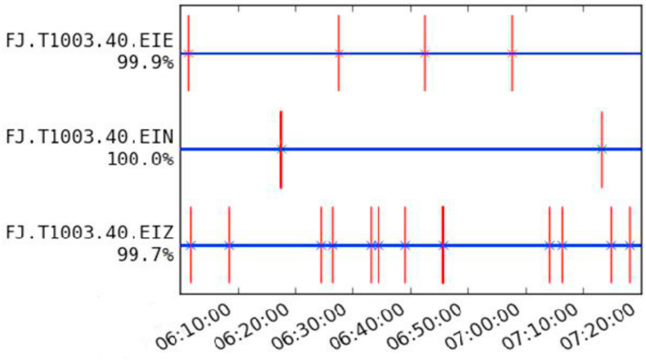

Fig. 7. The data receiving rate

3.6. Received signal strength

The Fig 8 indicates that the actual received values are close to the theoretical calculation ones. When the distance varies from 105km to 90km, the received signal strength of the system increases from -100dB to -80dB. When the distance is over 105km, the signal disappears.

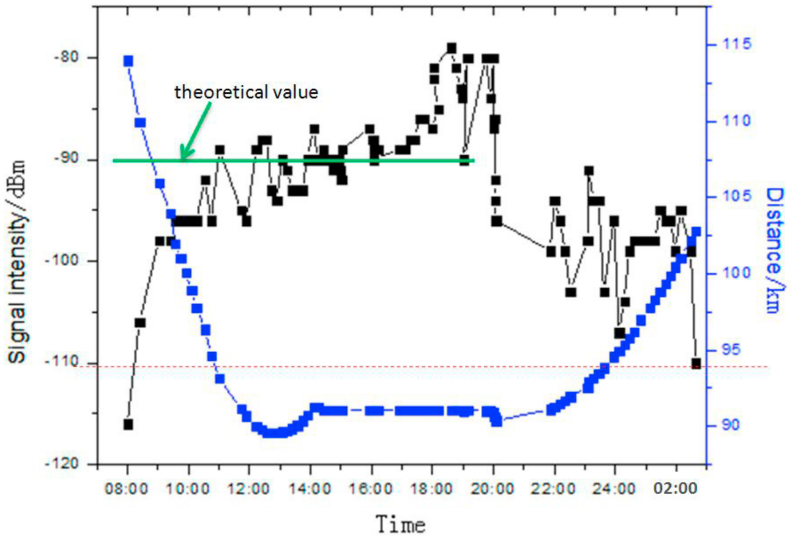

Fig. 8. The received signal strength

3.7. Data quality comparison

By comparison, it can be known that date transmitted by the seismic sensor T1003 at the maritime transmitting terminal through different channels to shore-based data receiving center are basically matched. As is shown inFig 9.

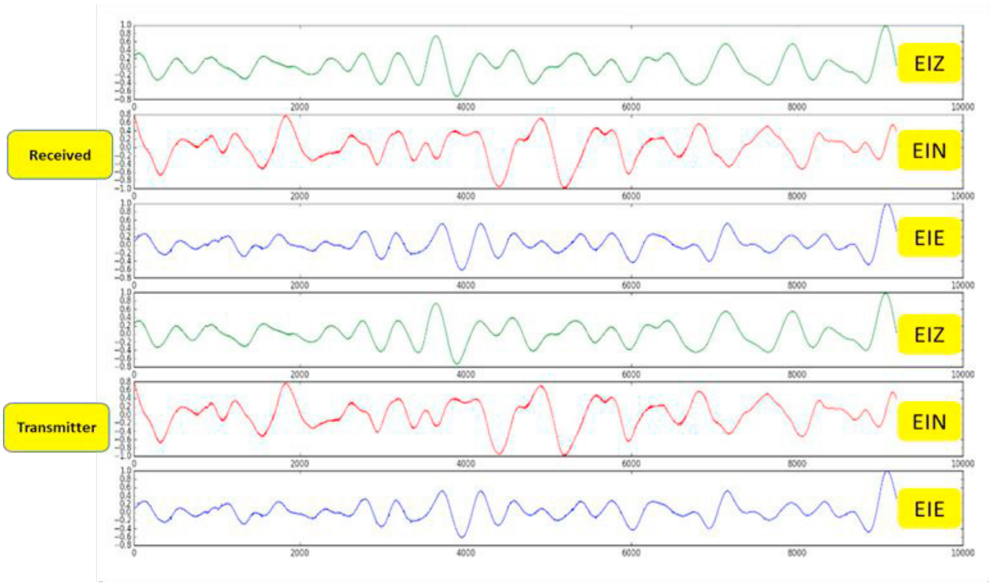

Fig. 9. The data quality comparison

4. Summary

The sea-area seismic wireless data transmission system has passed a series of sea-crossing experiments and several indicators of the whole system such as the transmission rate, transmission delay, and transmission loss rate, etc. have been tested through fixed-point transmission at sea. The stability performance, the data receiving performance and the receiving range limit of the system have all been verified, which proves the feasibility of the transmission link in the submarine seismic observation. In a word, this Marine seismic data transmission system can provide an experimental foundation for the national Marine seismic observation network.

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